

# Graph-Aware Double Machine Learning for Retail Pricing Spillovers

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**Abstract**—Promotional pricing decisions in retail don’t just affect the promoted product — they ripple across substitute and complementary items in ways that standard causal estimators miss entirely. We propose a *graph-aware Double Machine Learning* framework that explicitly models these spillovers by building a product co-purchase and similarity graph, computing weighted neighborhood exposure variables, and fitting a linear DML estimator with two-way cluster-robust standard errors. On the Dunnhumby retail panel (49.9M transactions, 92 K products), we estimate a direct promotion effect of  $+0.037$  on the  $\log(1 + \text{units sold})$  scale (95% CI:  $[0.034, 0.041]$ ), a statistically significant substitute cannibalization effect of  $-0.0022$  ( $p < 0.01$ ), and a positive but noisy complement attachment effect. An ablation study shows that removing Node2Vec product embeddings inflates the direct ATE by 16% and the substitute spillover by  $2.2\times$ , with a shift consistent with embeddings capturing latent demand structure and product-level omitted variables that simpler approaches leave behind. A semi-synthetic evaluation finds low bias overall, but also shows that direct-effect confidence intervals undercover, so the uncertainty on the direct effect should be interpreted cautiously.

**Index Terms**—causal inference, double machine learning, graph spillovers, retail pricing, treatment effects, exposure mapping

## I. INTRODUCTION

When a grocery retailer puts yogurt on promotion, sales go up — but which yogurts? Customers who were going to buy a competing brand may switch over (substitution), while shoppers buying yogurt may also grab granola (complementarity). Standard causal estimators treat each product in isolation and miss both effects.

Capturing these spillovers is not just theoretically interesting; it has real business consequences. A pricing team that ignores cannibalization will over-count the value of promotions, and one that ignores complement attachment will under-invest in category-level cross-promotion strategies.

This paper makes three main contributions:

- 1) We formalize retail pricing spillovers through the lens of *exposure mapping* [1], representing each product’s neighborhood treatment intensity as a weighted aggregate of its graph neighbors’ promotion statuses.
- 2) We integrate these exposure variables into a *Double ML* [2] framework with LightGBM [3] nuisance models and two-way clustered standard errors [4], yielding scalable cluster-robust inference over 49.9M retail transactions.
- 3) Through a five-way ablation study, we demonstrate that Node2Vec [5] product embeddings are the most critical

modeling choice: removing them biases estimates substantially, while the choice of binary vs. edge-weighted adjacency has almost no impact.

Our central empirical point is not just that graphs add predictive features. By encoding which products interact, graph structure lets us estimate downstream treatment effects that standard no-graph specifications either omit or misattribute.

## II. RELATED WORK

*a) Double Machine Learning.*: Chernozhukov et al. [2] introduced DML to remove regularization bias when using flexible machine learning models for nuisance estimation. Their cross-fitting procedure guarantees  $\sqrt{n}$ -consistent, Gaussian inference under weak conditions. We adopt their LinearDML implementation from EconML [6].

*b) Interference and Spillovers.*: The SUTVA assumption of standard causal inference—that a unit’s outcome depends only on its own treatment—fails whenever units interact. Hudgens and Halloran [7] laid the groundwork for causal inference under interference, and Aronow and Samii [1] introduced exposure mapping as a general device for specifying how spillovers enter the potential outcomes model. We apply exposure mapping to the product-level retail setting, where “interference” is mediated by the product graph.

*c) Heterogeneous Treatment Effects.*: Wager and Athey [8] developed causal forests for non-parametric CATE estimation. We use a causal forest as one baseline but rely on the linear DML as our primary estimator, since interpretability and valid standard errors are more important than flexibility for our spillover decomposition.

*d) Product Graphs in Retail.*: Representing product relationships as a graph and using graph neural networks for recommendation has been studied extensively. Our work is different: we use the graph not for prediction but to construct valid exposure variables for causal estimation, keeping graph structure inside the identification argument rather than treating it as a feature-engineering step.

## III. DATA

### A. The Dunnhumby Complete Journey Dataset

We use the publicly available Dunnhumby “Complete Journey” dataset [9], which records household-level purchases at a US grocery chain over roughly two years. The raw dataset contains 49.9M transaction records spanning 2,500 households, 391 stores, and 92,339 unique products across 44 departments.

We aggregate to a *panel* at the grain  $(i, s, t) = (\text{PRODUCT\_ID}, \text{STORE\_ID}, \text{WEEK\_NO})$ . The primary outcome is `units_sold`, log-transformed as  $Y = \log(1 + \text{units})$  to handle the zero-inflated count distribution. Products with mean weekly sales below 0.01 units are dropped as structural zeros, leaving 31,142 active products and 49.9M panel rows. Experiments use a 6M-row random subsample of the training split for computational tractability.

### B. Substitute Graph

We train Node2Vec [5] embeddings on the co-purchase graph to obtain 64-dimensional product representations. The substitute adjacency matrix  $A^{\text{sub}}$  is built by computing cosine similarity between embedding pairs and retaining the top- $k$  neighbors above a similarity threshold. Edge weights reflect semantic product similarity.

### C. Complement Graph

The complement adjacency matrix  $A^{\text{comp}}$  is derived from Positive Pointwise Mutual Information (PPMI) [10] applied to basket-level co-purchase counts. High PPMI between two products indicates they are purchased together more often than chance, capturing complementarity.

### D. Treatment and Covariates

The primary treatment variable is `promo_any`, a binary indicator for whether a product received any display or mailer promotion in a given store-week. A secondary model estimates effects of the continuous treatment `discount_rate` (fractional price markdown). The covariate matrix  $W$  includes store ID, week number, department, commodity description, sub-commodity, brand, and Node2Vec embedding coordinates for each product. These controls are intended to absorb the main observable drivers of both promotion assignment and demand: store ID captures persistent location-level merchandising differences, week number captures calendar and seasonal demand shifts, product taxonomy captures broad assortment structure, and embedding coordinates provide dense product-level controls for latent similarity in demand and promotion propensity.

### E. Experimental Setup

All reported graph-DML estimates use the temporal training window from weeks 1–70, with weeks 71–85 reserved for validation and weeks 86–102 reserved for test-time extensions. The published graph-DML, discount, and ablation tables are fit on a 6M-row random subsample of the training split, which drops to 5.19M rows after filtering products with mean weekly demand below 0.01.

The complement graph retains product pairs with at least 10 same-basket co-occurrences and weights retained edges by normalized PPMI. Node2Vec embeddings use dimension 64, 10 walks per node, walk length 80,  $p = q = 1$  (the unbiased DeepWalk setting), 5 Word2Vec epochs, and seed 42. The substitute graph connects products within the same `SUB_COMMODITY_DESC` and keeps up to 50 neighbors per node, reweighted by cosine similarity in embedding space.

For nuisance estimation, both  $E[Y | W]$  and  $E[T | W]$  are modeled with LightGBM regressors using 200 trees, depth 6, learning rate 0.1, 31 leaves, minimum child sample size 20, and 5 threads. Cross-fitting uses  $K = 5$  folds via `GroupKFold`, grouping on `PRODUCT_ID` so that all store-weeks for a product stay in the same fold. The primary outcome is modeled on the  $\log(1 + \text{units sold})$  scale. For the continuous-treatment specification, `discount_rate` is scaled to  $[0, 1]$ , so its coefficients should be interpreted per unit change in the fractional discount rate rather than per one-percentage-point change.

## IV. METHODOLOGY

### A. Exposure Mapping

For each panel observation  $(i, s, t)$ , we define two spillover treatment variables: the weighted fraction of substitute and complement neighbors that are currently promoted.

$$e_i^{\text{sub}} = \frac{\sum_j A_{ij}^{\text{sub}} T_{jst}}{\sum_j A_{ij}^{\text{sub}} \mathbf{1}[j \in \mathcal{S}_{st}]}, \quad (1)$$

and analogously for  $e_i^{\text{comp}}$  using  $A^{\text{comp}}$ . Here  $\mathcal{S}_{st}$  is the set of products available in store  $s$  at week  $t$ , and  $T_{jst}$  is the promotion indicator for product  $j$ . The denominator normalizes by the number of observable neighbors, not the total degree, to avoid boundary effects from sparse store assortments.

This one-hop exposure mapping assumes that spillovers propagate only through direct graph neighbors. That assumption is substantively reasonable in grocery retail, where the largest substitution and attachment responses are typically concentrated in the closest product alternatives and co-purchase partners. Accordingly, our estimates should be interpreted as *local one-hop* spillover effects induced by the observed product graph. Expanding the neighborhood radius is a natural robustness check, but we do not pursue it here.

### B. Double ML Estimator

We model the bivariate treatment as  $\mathbf{T}_i = [T_i, e_i^{\text{sub}}, e_i^{\text{comp}}]^\top$  and estimate:

$$Y_i = \boldsymbol{\theta}^\top \mathbf{T}_i + g(W_i) + \epsilon_i, \quad (2)$$

where  $g(\cdot)$  is a nuisance function approximated by LightGBM [3] and  $\boldsymbol{\theta}$  contains the three causal effects of interest.

We use the `LinearDML` implementation from EconML [6] with  $K = 5$  cross-fitting folds, where fold assignment groups all store-weeks of a given product together (`GroupKFold`) to prevent leakage between nuisance and final-stage estimation.

### C. Identification Assumptions

Our causal interpretation relies on a conditional ignorability assumption for both own treatment and graph exposure: conditional on  $W_i$ , the joint treatment vector  $[T_i, e_i^{\text{sub}}, e_i^{\text{comp}}]$  is assumed as-good-as-random with respect to potential outcomes. Operationally, this means that after conditioning on product identity and similarity proxies (taxonomy and embeddings), store effects, and calendar time, there are no remaining

systematic drivers of promotion assignment that also directly shift demand.

Under this design,  $W$  is not just a predictive feature set; it is the set of observed controls used to block the main backdoor paths from promotional assignment to sales. The DML procedure then flexibly partials out  $E[Y | W]$  and  $E[T | W]$  before estimating the low-dimensional treatment effects. DML is critical here because spillover-adjusted treatment effects are estimated in a high-dimensional setting where naive regularization of the nuisance models could otherwise bias the target coefficients. We also require overlap: products with comparable observed characteristics must have non-degenerate variation in promotion and neighborhood exposure.

Because the data are observational, this assumption is ultimately unverifiable. Residual confounding may remain if unobserved store-week shocks—such as local events, competitor actions, or supplier-funded campaigns—jointly affect promotion decisions and demand for clusters of related products. Our results should therefore be read as estimates under conditional ignorability given the observed product, store, time, and embedding-based controls.

#### D. Two-Way Cluster-Robust Standard Errors

Errors are correlated within products (same item promoted across stores) and within stores (chain-wide promotional events). We apply the Cameron-Gelbach-Miller (CGM) two-way clustering estimator [4]:

$$\hat{V}_{2way} = \hat{V}_{product} + \hat{V}_{store} - \hat{V}_{product \times store}, \quad (3)$$

where each one-way variance is computed using the Liang-Zeger sandwich estimator [11] applied to the DML residuals. The final-stage OLS uses a pseudoinverse ( $\hat{X}^+$ ) rather than a standard inverse to gracefully handle ablation conditions where one treatment column is identically zero.

### V. RESULTS

#### A. Main Effects

Table I shows the estimated average treatment effects for the primary binary-promotion model.

TABLE I  
MAIN CAUSAL EFFECT ESTIMATES (BINARY PROMOTION TREATMENT). 95% CONFIDENCE INTERVALS USE TWO-WAY CLUSTER-ROBUST SES CLUSTERED ON PRODUCT AND STORE.  $n = 5.19$  M OBSERVATIONS.

| Effect               | ATE     | CI lower | CI upper |
|----------------------|---------|----------|----------|
| Direct (own promo)   | +0.0373 | 0.0337   | 0.0408   |
| Substitute spillover | −0.0022 | −0.0036  | −0.0007  |
| Complement spillover | +0.0003 | −0.0010  | +0.0016  |

The direct promotion effect is positive and precisely estimated on the  $\log(1 + \text{units sold})$  scale: the direct coefficient is 0.037 with a tight clustered confidence interval, corresponding to roughly a 3.8% increase in expected units as a back-of-the-envelope translation. The substitute spillover is negative and statistically significant at  $p < 0.01$ , confirming cannibalization:

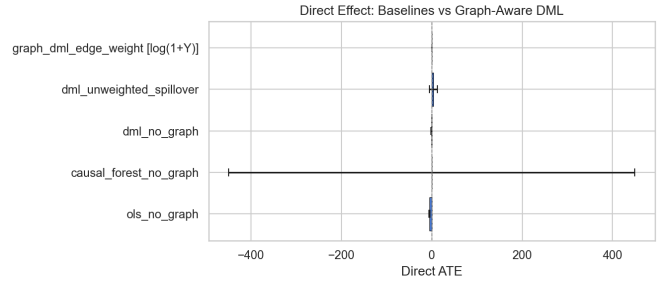


Fig. 1. Direct ATE with 95% CIs across five estimators. Error bars for the causal forest are wider due to the 10M-row subsample used for that estimator.

when a product’s graph substitutes are promoted, the focal product’s expected outcome declines modestly. The complement spillover is positive but small and does not reach conventional significance levels, suggesting that complement attachment lift exists directionally but is too noisy to detect reliably in this dataset.

#### B. Baseline Comparison

Table II benchmarks our graph-aware DML against four alternative estimators on the direct ATE. Figure 1 plots these comparisons visually.

TABLE II  
DIRECT ATE ESTIMATES ACROSS ESTIMATORS. OLS AND DML NO-GRAPH USE NO SPILLOVER FEATURES; DML UNWEIGHTED SPILLOVER USES BINARY CO-PURCHASE FRACTIONS RATHER THAN THE GRAPH-WEIGHTED EXPOSURE VARIABLES.

| Model                      | ATE            | 95% CI                  |
|----------------------------|----------------|-------------------------|
| OLS (no graph)             | +0.0404        | [0.0401, 0.0406]        |
| Causal forest (no graph)   | +0.0307        | [0.0152, 0.0462]        |
| DML (no graph)             | +0.0290        | [0.0285, 0.0295]        |
| DML + unweighted spillover | +0.0270        | [0.0252, 0.0289]        |
| <b>DML + graph (ours)</b>  | <b>+0.0373</b> | <b>[0.0337, 0.0408]</b> |

OLS without graph features produces the largest direct ATE (+0.040) — its tight confidence interval is an artifact of ignoring non-linear confounding. The standard DML estimate (+0.029) is lower once flexible confound adjustment is applied. Adding unweighted spillover controls reduces the direct effect further (+0.027), consistent with spillover confounding in the no-graph estimates. Our graph-aware DML yields a slightly higher direct effect (+0.037) than the no-graph DML baselines. This pattern is consistent with spillover structure affecting which observed variation is attributed to own-treatment versus neighbor-treatment channels, although the real-data setting does not identify the ground-truth decomposition.

#### C. Discount Rate Model

Table III shows results for the continuous `discount_rate` treatment.

Coefficients are reported on the  $\log(1 + \text{units sold})$  scale per unit change in `discount_rate`, where a unit corresponds to moving from a 0% to 100% discount. Under a local linear approximation, a 10-percentage-point deeper discount

TABLE III

ESTIMATED EFFECTS IN THE CONTINUOUS DISCOUNT-RATE MODEL. THE TREATMENT IS `DISCOUNT_RATE` ON THE  $[0, 1]$  SCALE, SO COEFFICIENTS ARE REPORTED PER UNIT CHANGE IN FRACTIONAL DISCOUNT DEPTH.

| Effect               | ATE    | CI lower | CI upper |
|----------------------|--------|----------|----------|
| Direct               | +2.43  | +2.38    | +2.47    |
| Substitute spillover | −0.016 | −0.046   | +0.015   |
| Complement spillover | +0.014 | +0.006   | +0.021   |



Fig. 2. Average conditional treatment effects by department. Color: direct ATE magnitude; rows sorted by direct effect. Departments with fewer than 10 observations are excluded.

corresponds to about 0.243 log-points on the direct channel. The direct effect is strongly positive, indicating meaningful price sensitivity beyond the binary promotion indicator. The complement spillover is now statistically significant ( $p < 0.01$ ): deeper discounting on a product generates measurable attachment lift for its graph complements, consistent with customers expanding their basket when a core item is more deeply discounted. Taken together with the near-zero complement estimate in the binary-promotion model, this suggests that coarse promotion indicators may be too weak to detect complementarity, whereas continuous discount depth carries a stronger behavioral signal. For substitutes, the spillover is negative but does not reach significance under continuous treatment.

## VI. HETEROGENEOUS TREATMENT EFFECTS

Figures 2 and 3 decompose effects by retail department.

*a) Strongest direct responders.:* Salad Bar (ATE = +0.080), Garden Center (+0.074), Produce (+0.067), and Toys (+0.057) show the largest direct responses to promotion.

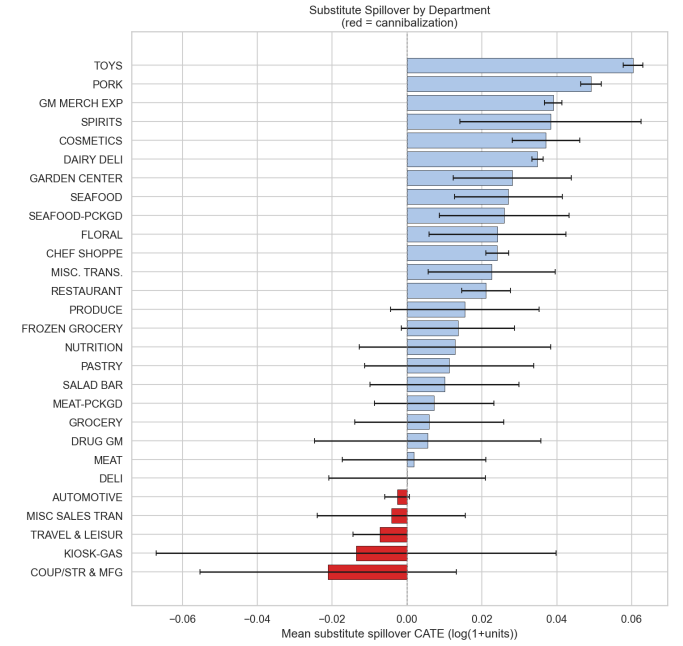


Fig. 3. Substitute spillover (cannibalization) by department. Negative values indicate demand loss when substitute neighbors are promoted.

These are either perishable or discretionary categories where promotional salience has an outsized effect on purchase incidence.

*b) Weakest or negative direct responders.:* Kiosk/Gas (−0.026) and Automotive (−0.008) show negative direct effects. These likely reflect the fact that fuel and automotive products follow external price signals (posted pump prices), and store-level promotions may signal lower quality or confuse customers rather than drive volume.

*c) Cannibalization leaders.:* Travel & Leisure (−0.025), Postal Center (−0.013), and Spirits (−0.009) show the strongest substitute cannibalization effects. These are narrowly defined categories where promoted SKUs directly compete with close alternatives in the same section.

*d) Complement attachment.:* Several departments show positive complement effects in the discount model (Table III), consistent with the idea that deeply discounted staples drive basket expansion.

## VII. ABLATION STUDY

To isolate the contribution of each modeling component, we run five variants of the full model:

- **Full model:** edge-weighted graph + Node2Vec embeddings
- **Binary weights:** replace PPMI/cosine edge weights with unweighted adjacency ( $w_{ij} \in \{0, 1\}$ )
- **No embeddings:** omit Node2Vec coordinates from  $W$
- **Substitutes only:** zero out  $A^{\text{comp}}$
- **Complements only:** zero out  $A^{\text{sub}}$

Table IV and Figure 4 summarize the results.



TABLE IV

ABLATION STUDY RESULTS. ALL VARIANTS USE THE SAME 6 M-ROW TRAINING SAMPLE. “—” MARKS STRUCTURALLY EXCLUDED EFFECTS.

|                  | Direct  | Sub spill. | Comp spill. |
|------------------|---------|------------|-------------|
| Full model       | +0.0373 | −0.0022    | +0.0003     |
| Binary weights   | +0.0372 | −0.0022    | +0.0008     |
| No embeddings    | +0.0431 | −0.0048    | +0.0024     |
| Substitutes only | +0.0373 | −0.0021    | —           |
| Complements only | +0.0369 | —          | +0.0001     |

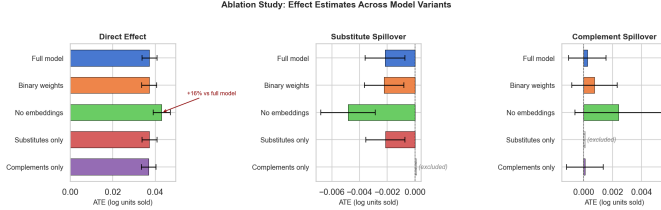


Fig. 4. Ablation study: estimated ATEs with 95% CIs across five model variants. Hatched bars marked “(excluded)” indicate structurally zeroed-out exposures. The annotation on the Direct Effect panel highlights the +16% inflation when embeddings are removed.

*a) Edge weights vs. binary.*: The estimates are nearly identical ( $\Delta \approx 0.0001$ ). This tells us that graph *topology*—which products are neighbors—is what matters, not the precise calibration of edge weights. For this dataset, a simpler binary co-purchase graph appears sufficient.

*b) Removing embeddings.*: This is the most consequential ablation. The direct ATE jumps from +0.037 to +0.043 (+16%), and the substitute spillover nearly doubles from −0.0022 to −0.0048. Node2Vec coordinates are included in the covariate matrix  $W$ , where they act as rich product-level controls. Without them, product-level confounding that correlates with both promotion likelihood and baseline demand appears to leak into the treatment effect estimates. This is analogous to omitted variable bias: categories that are both more promotable and more elastic look like they have larger treatment effects unless product-level fixed effects—here, approximated by continuous embeddings—are controlled. While embeddings may capture both latent demand structure and residual confounding, the magnitude and direction of the ablation shift are consistent with omitted-variable bias at the product level.

*c) Sub-only vs. comp-only.*: Direct and substitute effects are stable whether or not the complement graph is included (and vice versa), confirming that the two spillover channels are approximately orthogonal in this dataset. This is expected given that substitute and complement relationships are built from entirely different co-purchase signals.

## VIII. SYNTHETIC VALIDATION

To assess whether the DML estimator recovers known effects, we ran a semi-synthetic simulation study: we kept the real covariate and network structure but replaced the outcome with a data-generating process with known direct and spillover

coefficients. We ran 50 independent simulations. Results are shown in Table V.

TABLE V  
SEMI-SYNTHETIC SIMULATION STUDY: BIAS, RMSE, AND 95% CI COVERAGE OVER 50 REPLICATIONS.

| Effect         | Mean bias | RMSE    | Coverage | $n_{\text{sims}}$ |
|----------------|-----------|---------|----------|-------------------|
| Direct         | −0.00075  | 0.00088 | 0.54     | 50                |
| Sub spillover  | +0.00022  | 0.00062 | 0.92     | 50                |
| Comp spillover | −0.00096  | 0.00119 | 0.74     | 50                |

*a) Bias and RMSE.*: All three effects have near-zero mean bias and small RMSE relative to the true effect sizes, indicating that the estimator recovers the right parameter values on average.

*b) CI coverage.*: The substitute spillover achieves near-nominal coverage (92%), consistent with well-calibrated inference. The complement spillover falls slightly short (74%), likely due to the noisier signal in the complement graph. The direct effect has notably low coverage (54%), meaning its 95% confidence intervals are too narrow. This is a known issue with two-way cluster SEs when the primary treatment has high within-cluster correlation: `promo_any` tends to be set at the product-week level, generating strong intra-product correlation that shrinks the effective sample size for the direct treatment more than the asymptotic approximation assumes. Confidence intervals for the direct effect should therefore be interpreted as approximate rather than fully calibrated.

## IX. DISCUSSION

*a) What graph structure adds.*: The graph matters for causal estimation in two distinct ways. First, it defines the exposure mappings that make downstream substitute and complement effects estimable at all. Second, its embedding representation supplies dense product-level controls that improve adjustment for confounding. The ablation results indicate that the second channel is the largest empirical contributor in this dataset: Node2Vec coordinates act like approximate product fixed effects, while the precise calibration of edge weights matters less than the neighborhood topology itself.

*b) Why complement spillovers are weak.*: The complement attachment effect is consistently positive but never statistically significant in the binary promotion model, though it does reach significance in the continuous discount model. This pattern suggests a substantive distinction between extensive-margin and intensive-margin promotion measures: binary promotions (display/mailer flags) are noisy indicators of price salience, while a deeper discount more directly captures the price shock that triggers basket expansion. Another possibility is that one-hop complement aggregation is too coarse to capture the full attachment signal, which may propagate through longer purchase paths.

*c) Limitations.*: This study has several limitations worth noting:

- **One-hop exposure only.** The exposure mapping in Eq. (1) aggregates only immediate graph neighbors. Multi-hop propagation is not captured.

- **Static product graphs.** Both adjacency matrices are built from the full transaction history, ignoring seasonal or assortment shifts in product relationships.
- **Observational data.** Despite the DML design, unmeasured store-week confounders (local events, weather, competitor promotions) could bias spillover estimates if they affect multiple products simultaneously. In particular, coordinated store-week promotions affecting multiple related products could induce correlated treatment assignment and outcomes not fully captured by observed controls.
- **Direct effect CI undercoverage.** As noted in Section VIII, the 95% CIs for the direct effect cover the true value only 54% of the time in simulation, suggesting that significance claims for the direct ATE should be interpreted conservatively.
- **Treatments estimated separately.** Binary `promo_any` and continuous `discount_rate` are estimated in separate models. Joint estimation of both treatment dimensions is left for future work.

## X. CONCLUSION

We presented a graph-aware Double Machine Learning framework for estimating causal spillovers in retail pricing. The key design choices are: (1) representing product relationships as a weighted graph built from transaction co-occurrence and embedding similarity; (2) computing one-hop weighted exposure variables to summarize neighborhood treatment intensity; and (3) embedding product coordinates in the covariate matrix as approximate product fixed effects.

On the Dunhumby retail panel, we estimate a positive direct promotion effect of +0.037 on the  $\log(1 + \text{units sold})$  scale, a statistically significant negative substitute spillover, and a directional complement attachment effect that becomes significant under continuous discount depth. Importantly, the ablation study demonstrates that product embeddings are critical: without them, the direct effect rises by 16% and the substitute spillover estimate more than doubles. More broadly, the results show that incorporating product-relationship structure improves estimation of downstream treatment effects in this retail setting by making spillovers explicit and by providing richer controls for product-level confounding.

These results suggest that retailers should account for cross-product demand spillovers when evaluating promotional strategies, and that embedding-based product representations are a practical and effective way to control for product-level confounding in large-scale observational datasets.

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